

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

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[@den-run-ai](#) · github.com/den-run-ai/askme

AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one JSON action → execution evidence ↻

01

CONTROLLABLE SMALL LLMS

Choose the model boundary

- Execution speed
- Hardware and deployment
- Post-training access



02

ASKME LOOP

One small action at a time

```
plan → act → execute / test  
→ fresh evidence ↻
```

Interpret the result. Update locally when possible.



03

EXTERNAL ACCEPTANCE LAYER

Check the delivered workflow

- Preserve the full contract
- Check required behavior
- Accept the real artifact independently

The bridge: fixed action vocabulary + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

The retained record does not independently prove stdout or source contents.

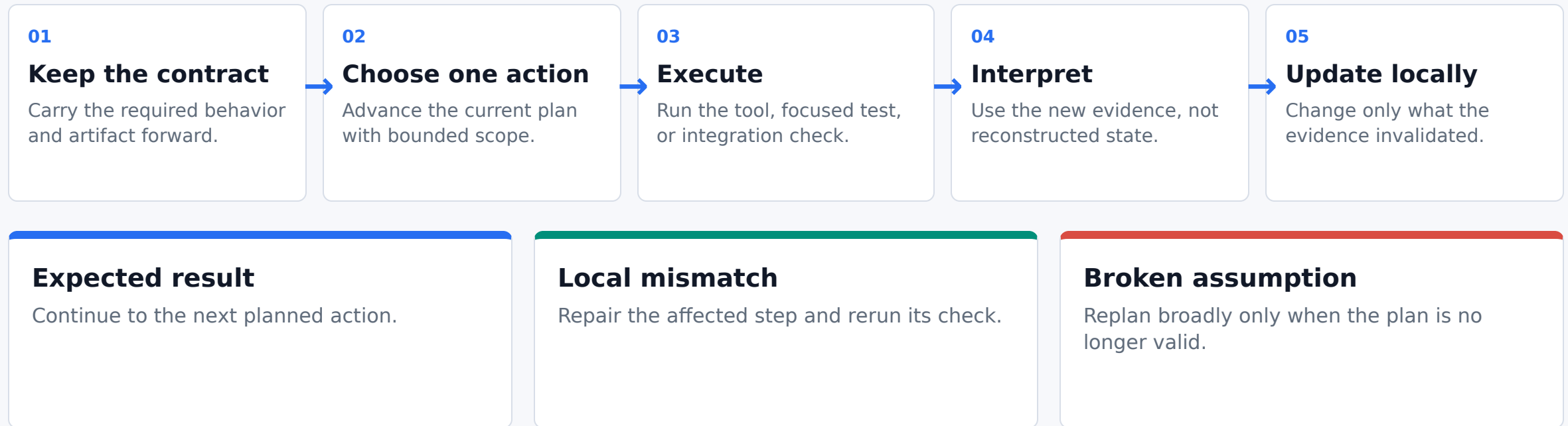
Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion $\neq$ accepted artifact.

Current limit: acceptance scored this run after completion; the failure was not returned for recovery.

Hypothesis: update only what fresh evidence invalidates



Trajectory goal—not yet isolated: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

All four variants ran; one deliverable was rejected

4 hosted variants × 2 deliberately simple checks × 1 unseeded trajectory per cell

8 / 8 reported complete

7 / 8 artifact accepted

Compatibility smoke · n=1/cell · descriptive only

MODEL · SHAPE	ARTIFACT BUILD · OBSERVED TRAJECTORY	SANITY REPAIR · OBSERVED TRAJECTORY
Gemma 4 26B A4B ◆ MoE · 25.2B total / 3.8B active	ACCEPTED 603.6s · 19 steps · 19.5k tok 1 full / 1 local replan	ACCEPTED 20.0s · 7 steps · 4.5k tok 0 full / 0 local replans
Gemma 4 31B ● Dense · 30.7B parameters	ACCEPTED 66.5s · 4 steps · 4.4k tok 0 full / 0 local replans	ACCEPTED 22.4s · 5 steps · 4.2k tok 0 full / 1 local replan
Qwen3.6-27B ● Dense · 27B parameters	ACCEPTED 47.9s · 6 steps · 5.3k tok 0 full / 0 local replans	ACCEPTED 23.0s · 6 steps · 4.1k tok 0 full / 0 local replans
Qwen3.6-35B-A3B ◆ MoE · 35B total / 3B active	NOT ACCEPTED 17.7s · 3 steps · 3.5k tok wrong artifact path	ACCEPTED 11.8s · 4 steps · 2.5k tok 0 full / 0 local replans

Supported
All four reported completion on both checks; acceptance rejected one wrong deliverable.

Not a clean model comparison
No pair isolates size: dense/MoE shape, active compute, run order, and trajectories changed. No Qwen-vs-Gemma, larger-vs-smaller, speed, reasoning, or reliability inference.

Feedback works only when actions enter and failures return

One limit appeared before execution; another after reported completion.

01 · ENTER THE LOOP

The model must emit a valid action

model → valid structured action

FeatureBench-fast · 1 task · Gemma 4 31B:

3 plans → 4 reads → 0 writes → empty patch → unresolved.

02 · INSIDE THE LOOP

Execution feedback can guide repair

action → execute / test → evidence ↺

AskMe can continue, update locally, or replan.

03 · AFTER COMPLETION

Acceptance checks the deliverable

artifact → independent acceptance

The Qwen wrong-path result was rejected, but not returned for recovery.

External boundary probe—not a score: the FeatureBench canary exposed a 512-token structured-action bottleneck before a patch existed. One task cannot separate model capability from this model-harness interface.

Requalified Aug 1 · one task, one attempt/model: under the revision-3 bundle and a changed serving stack, both patches applied but remained unresolved — Gemma 11/13 target tests (same two failures as the exploratory pi run); Qwen 7/13 (pi: 10/13). Gemma rewrote 18 times without testing; Qwen wrote once, then resumed reading; neither emitted `done`.

Promising for bounded loops. Feature readiness is still open.

Observed

Simple smoke: 7 / 8 artifacts accepted. On one feature task, revision 3 moved both models from empty to applied but unresolved patches: Gemma 11/13 target tests; Qwen 7/13.

Supported

Harness design was consequential on this task. The bundled changes moved failures into execution: Gemma rewrote without testing; Qwen resumed observation after one write.

Still open

Reliability beyond one task; clean validation and termination; transport versus serving/configuration effects; reasoning, family, size, architecture, and local performance. Revision 4 is implemented in open PR #21; v7 requalification is pending.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries—not a leaderboard.

	AskMe	pi	OpenHands
ACTION SURFACE	6 fixed JSON actions; exactly one action per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible <code>Action → Observation</code> tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	<code>done</code> → conditional fail-open validation; held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	<code>finish</code> signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.

Sources: AskMe architecture · [pi coding-agent docs](#) · [OpenHands tool system](#) and [conversation architecture](#)